

NeuroTechX

NeuroTechX: Building a Community

Our work, our contribution, and the next neurotech founders

Morgan Hough
Executive Director, NeuroTechX

Japan BMI/BCI Ecosystem Kick-off
Coral Capital, Tokyo / Hybrid
September 7, 2026

2026-09-04

NeuroTechX / Community

[0:00–1:00] Introduce yourself as Morgan Hough, Executive Director of NeuroTechX. Thank the organizers. This talk is about NeuroTechX: what it is, what we do, how that work contributes to the community, and what it can mean for people building neurotech companies. The Tokyo meeting is the audience context.

NeuroTechX

**NeuroTechX:
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What is NeuroTechX?

A nonprofit community

Bringing together people interested in neuroscience and technology.

Local beginnings, international connections

Founded in Montréal in 2015; connected through city chapters, student groups, and open projects.

Three pillars

Community, education, and professional development.

NeuroTechX: Our Story

2026-09-04

NeuroTechX / Community

└─ What is NeuroTechX?

└─ What is NeuroTechX?

[1:00–2:30] Your original deck describes the organization as originally Societe BCI de Mon-treal and then a global community. The public history dates NeuroTechX's founding to 2015. Introduce the purpose without membership totals or superlatives. The exact legal history is not expanded beyond your supplied account.

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NeuroTechX: Our Story

What do we do?

Bring people together

Chapter meetings, hacknights, student communities, and hackathons.

Make learning accessible

The NeuroTech Primer, workshops, webinars, and practical experiments.

Build and share tools

EEG-ExPy, MOABB, and contributions across the open neurotechnology ecosystem.

NeuroTechX: education and developer resources; Morgan's original presentation

2026-09-04

NeuroTechX / Community

└─ What is NeuroTechX?

└─ What do we do?

[2:30–3:30] This is the overview of the activities in your original slides. The next section makes each concrete. Be careful to distinguish NeuroTechX projects from independent tools our community uses or contributes to.

What do we do?

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NeuroTechX: education and developer resources; Morgan's original presentation

NeuroTechSF: the weekly hacknight

A recurring place to meet

Newcomers welcome alongside people already working on projects.

In-person and remote participation

Our chapter's format includes online participation, livestreaming, and recordings.

Something to work on together

Hardware for hacking, experiments, software, and help from peers.

NeuroTechSF events; format described in Morgan's original slides

2026-09-04

NeuroTechX / Community

└─What do we do?

└─NeuroTechSF: the weekly hacknight

[3:30–5:00] Use your experience of the SF hacknight. The original deck describes Wednesday meetings, Jitsi participation, livestreaming, YouTube archives, and hardware access. We omit the exact weekly time because this is a historical account rather than a current event listing. Explain what a newcomer actually encounters. No retention rate or success story has been inferred.

NeuroTechSF: the weekly hacknight

A recurring place to meet
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Hardware for hacking, experiments, software, and help from peers.

NeuroTechSF events; format described in Morgan's original slides

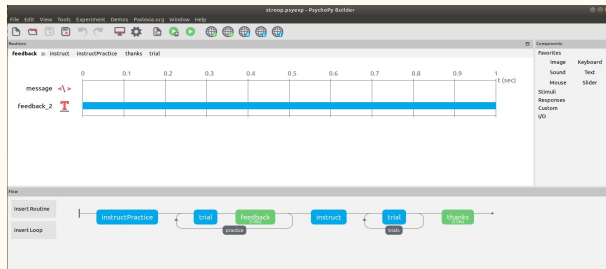
EEG-ExPy: a shared experiment

EEG experiments in Python

A project people can run, inspect, and help improve.

Experiment design connects with software and data collection.

EEG-ExPy; experiment illustration from Morgan's original slide 10



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└─What do we do?

└─EEG-ExPy: a shared experiment

[5:00–6:30] Explain EEG-ExPy using your original material. The screenshot is the PsychoPy experiment interface from your deck, used as an illustration of experiment design. It is not presented as a new EEG-ExPy interface or validation result. Avoid repeating the old deck's broad platform and headset compatibility list without version-specific testing.

EEG-ExPy: a shared experiment

EEG experiments in Python

A project people can run, inspect, and help improve.

Experiment design connects with software and data collection.

EEG-ExPy; experiment illustration from Morgan's original slide 10

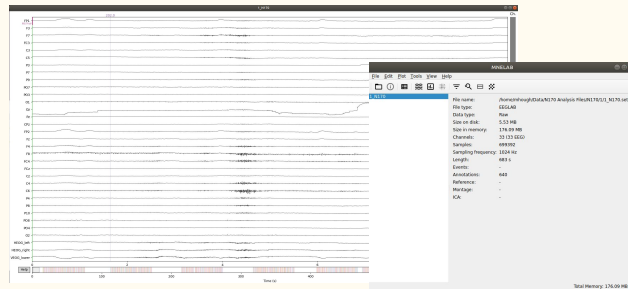


Citizen science: from recording to analysis

Participation can include the experiment itself.

Our examples include N170 experiments and resting-state recordings.

Looking at the signal is part of learning how to collect it.



Morgan's original presentation: citizen science and EEG analysis

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└ What do we do?

└ Citizen science: from recording to analysis

[6:30–8:00] Describe the citizen-science work from your own experience. The figure is your original N170 recording and analysis screenshot. It shows inspection of data, not evidence of a population-level finding or a validated application. The old slide's proposed uploads and cloud support are not represented as services currently promised by NeuroTechX.

Citizen science: from recording to analysis

Participation can include the experiment itself. Our examples include N170 experiments and resting-state recordings. Looking at the signal is part of learning how to collect it.



Morgan's original presentation: citizen science and EEG analysis

MOABB: making BCI comparisons reproducible

Mother of All BCI Benchmarks

An open framework for comparing algorithms on public EEG datasets.

A concrete research contribution

The 2024 benchmark study evaluated 30 machine-learning pipelines across 36 datasets, covering motor imagery, P300, and SSVEP.

Shared implementations and evaluation methods give other researchers a starting point they can inspect and reproduce.

Chevallier et al., 2024, arXiv:2404.15319

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└─What do we do?

└─MOABB: making BCI comparisons reproducible

[8:00–10:00] MOABB is an example of community work producing a research artifact beyond a single event. The numerical scope is specific to the cited 2024 study, not today’s dataset total. Credit the contributors. Do not generalize the study into a claim that a particular algorithm always wins or that benchmark performance establishes clinical utility.

MOABB: making BCI comparisons reproducible

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Education: a way into the field

The NeuroTech Primer

An introduction to neurotechnology, including recording and stimulation.

Workshops and webinars

Opportunities to learn methods and hear from people working in the field.

Learning through contribution

Developer meetings and practical projects alongside written resources.

NeuroTechX: The NeuroTech Primer, learning resources, and developer meetings

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NeuroTechX / Community

└─ What do we do?

└─ Education: a way into the field

[10:00–11:30] Connect this to the book and education slides in your original deck. These are several routes into participation, without claiming that every learner follows the same path. Not every resource linked on the education page was created by NeuroTechX.

Education: a way into the field

The NeuroTech Primer

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NeuroTechX: The NeuroTech Primer, learning resources, and developer meetings

What these activities contribute

Access to people

Meetings and events create opportunities to ask questions and find collaborators.

Access to practice

Shared experiments and workshops give people something concrete to try.

Work others can reuse

Code, benchmarks, recordings, and learning materials remain available beyond a meeting.

Synthesis of the activities and resources shown in this talk

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NeuroTechX / Community

└─ How do we add to the community?

└─ What these activities contribute

[11:30–13:00] This is a synthesis of the documented activities, not a claim about measured career or startup outcomes. Use a firsthand example if you want to illustrate the value to a particular person. The distinction is between what we provide and what a participant chooses to do with it.

What these activities contribute

Access to people

Meetings and events create opportunities to ask questions and find collaborators.

Access to practice

Shared experiments and workshops give people something concrete to try.

Work others can reuse

Code, benchmarks, recordings, and learning materials remain available beyond a meeting.

Synthesis of the activities and resources shown in this talk

The global hackathon: 2022

October 29–30, 2022

Global online participation, with local chapter hosting.
Free and open to the public.

Five listed in-person locations

Paris, Barcelona, Belgrade, San Francisco, and Munich.

Three ways to participate

Hands-on technology; an EEG data challenge;
design, strategy, and other disciplines.

neurotechx.com/hackathon2022

2026-09-04

NeuroTechX / Community

└─ How do we add to the community?

└─ The global hackathon: 2022

Historical context, approximately one minute. Morgan identifies the 2022 event as historically important. The original event page describes the format, dates, listed chapter hosts, and tracks. Its description calls this the first annual global NeuroTechX hackathon; this does not establish that it was NeuroTechX's first hackathon of any kind. The data challenge description credits Inria with an EEG dataset and infrastructure but contains leftover editing instructions. The slide summarizes the track without reproducing that malformed text. Listed locations are the published hosting plan, not verified attendance. Leave the account of why this event mattered, what worked, and what did not to Morgan.

The global hackathon: 2022

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Three ways to participate

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neurotechx.com/hackathon2022

The global hackathon: 2023

December 2–3, 2023

The second annual global event used a hybrid format.
Free and open to the public.

Eight listed in-person locations

Paris, Vienna, Zurich, Delhi, Kharagpur,
Brasília, Buenos Aires, and London.

Four tracks

Technology and hardware; data and algorithms;
design and strategy; a newly added ethics track.

neurotechx.com/hackathon2023

2026-09-04

NeuroTechX / Community

└─How do we add to the community?

└─The global hackathon: 2023

Historical context, approximately one minute. The event page identifies this as the second global annual NeuroTechX Hackathon. It lists eight hosting cities and explicitly describes the ethics track as an addition. These are changes to the published program, not evidence of increased actual attendance or improved outcomes. The page also describes team-formation assistance and organizing-team and mentor support; Morgan may wish to discuss how those worked in practice. Do not infer that all eight locations ran as advertised or that a larger city list proves community growth. No website screenshots, visual styling, sponsor graphics, or headshots are reused.

The global hackathon: 2023

December 2–3, 2023

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Paris, Vienna, Zurich, Delhi, Kharagpur,
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Technology and hardware; data and algorithms;
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neurotechx.com/hackathon2023

Global NeuroHack 2026

April 10–12, 2026

Frontier Tower, San Francisco

A recent event we hosted, with NeuroTechX helping organize and support the hackathon.

Three tracks:

- Blue-Sky Prototyping
- Non-Implantable Speech Decoding
- Maximizing Bit Rate

The program included workshops, mentor check-ins, and project presentations.

global-neurohack.github.io

2026-09-04

NeuroTechX / Community

└─ How do we add to the community?

└─ Global NeuroHack 2026

Historical context, approximately one minute. Morgan identified this as a recent event NeuroTechX hosted. The event website supplies the date, venue, tracks, and program, and credits NeuroTechX with organizing support. Its copy remains prospective: do not present advertised attendance, country counts, mentor counts, or scheduled activities as independently verified outcomes. Morgan's firsthand account should supply actual turnout, memorable projects, what worked, what did not, and any subsequent collaborations. No lesson or outcome has been invented here.

Global NeuroHack 2026

April 10–12, 2026

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Three tracks:

- Blue-Sky Prototyping
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global-neurohack.github.io

What the hackathon history shows

Different ways to join

The 2022 and 2023 programs included technical work, data, and design.

Different places to participate

The earlier global programs combined online participation with local hosts.

Different questions to work on

The 2023 program added ethics; the 2026 tracks included speech decoding and BCI games.

2022 and 2023 event programs; Global NeuroHack 2026 program

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NeuroTechX / Community

└─ How do we add to the community?

└─ What the hackathon history shows

[16:00–18:00] These are observable features of the programs already described. This is where your own lessons about hosting, access, team formation, or follow-through belong. The current text deliberately does not invent what participants learned, whether teams continued, or which format worked best. Explain the historical importance in your own words.

What the hackathon history shows

Different ways to join

The 2022 and 2023 programs included technical work, data, and design.

Different places to participate

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Different questions to work on

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2022 and 2023 event programs; Global NeuroHack 2026 program

Founders can build on work already shared

Learn the field

Use educational resources and community conversations to develop a working understanding.

Explore an idea

Use open experiments and tools to investigate a technical question.

Test the claim

Use reproducible baselines to ask what an approach actually adds.

Founder-facing implications of the resources shown; MOABB and EEG-ExPy

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NeuroTechX / Community

└─What does the future hold for neurotech founders?

└─Founders can build on work already shared

[18:00–20:00] This is the transition to the future, grounded in resources that exist. It is a suggested use of community resources, not a new NeuroTechX program or a promise of company formation. Open projects can inform exploration, but an intended product must establish its own requirements and evidence. No licensing or commercial-use advice is implied.

Founders can build on work already shared

Learn the field
Use educational resources and community conversations to develop a working understanding.
Explore an idea
Use open experiments and tools to investigate a technical question.
Test the claim
Use reproducible baselines to ask what an approach actually adds.

Founder-facing implications of the resources shown; MOABB and EEG-ExPy

Where could the next founder contribute?

Tools for people doing neurotechnology

Recording workflows, experiments, analysis, and reproducibility.

Applications built around a specific need

The recent hackathon explored speech interfaces, BCI games, and open-ended prototypes.

The practical questions

Who needs this? What works today? What evidence is still missing?

Global NeuroHack 2026 tracks; application questions for discussion

2026-09-04

NeuroTechX / Community

└─What does the future hold for neurotech founders?

└─Where could the next founder contribute?

[20:00–22:00] Present these as directions to explore, not predictions of market size, time-lines, or technical readiness. In particular, a non-implantable speech-decoding challenge does not establish a working speech product. A game built with mock neural data is not a demonstrated human BCI. Keep the distinction between a project prompt and a product claim clear.

Where could the next founder contribute?

Tools for people doing neurotechnology

Recording workflows, experiments, analysis, and reproducibility.

Applications built around a specific need

The recent hackathon explored speech interfaces, BCI games, and open-ended prototypes.

The practical questions

Who needs this? What works today? What evidence is still missing?

Global NeuroHack 2026 tracks; application questions for discussion

A founder can also be a community contributor

Bring a question people can work on

A technical problem, an experiment, or a result to examine together.

Share something useful

Documentation, a reproducible example, a workshop, or time helping another builder.

Build relationships through the work

Meet people by learning, making, and contributing alongside them.

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NeuroTechX / Community

└─What does the future hold for neurotech founders?

└─A founder can also be a community contributor

[22:00–24:00] This is an invitation to participate, grounded in the existing formats. It does not promise cofounder matching, funding, regulatory support, or access to a particular partner. Invite founders to think about what they can add to the community as well as what they hope to find. This is discussion framing, not an organizational commitment.

A founder can also be a community contributor

Bring a question people can work on

A technical problem, an experiment, or a result to examine together.

Share something useful

Documentation, a reproducible example, a workshop, or time helping another builder.

Build relationships through the work

Meet people by learning, making, and contributing alongside them.

What can we build and learn together?

Join the community

neurotechx.org/community

Explore the work

github.com/NeuroTechX

Find a NeuroTechSF event

lu.ma/neurotechsf

Morgan Hough

Executive Director, NeuroTechX

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NeuroTechX / Community

└─ What does the future hold for neurotech founders?

└─ What can we build and learn together?

[24:00–25:00] Thank the audience. Return to NeuroTechX as a community people can join and contribute to. Invite questions about the history, activities, and what founders need from a community. [25:00–30:00] Discussion.

What can we build and learn together?

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neurotechx.org/community

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github.com/NeuroTechX

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Sources

NeuroTechX and its activities

Organization history Education and developer resources

Projects

EEG-ExPy MOABB 2024 MOABB benchmark study

Hackathon history

2022 2023 Global NeuroHack 2026

Firsthand presentation material

Morgan’s original NeuroTechX presentation

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Sources

Backup only. SOURCES.md maps facts and figures to their origins. The future section is discussion grounded in existing resources, not a forecast or an announcement.

Sources

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